Test success (%) Mean ± sample SD (%)

Method	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
	AppWorld	PlanBench-XL	AppWorld	PlanBench-XL	AppWorld	PlanBench-XL
Frozen Base	34.92 ±1.37	35.93 ±3.90	36.90 ±1.57	36.67 ±4.44	46.03 ±1.37	45.56 ±5.56
AREW / T3	37.70 ±1.72	38.52 ±2.80	38.49 ±1.24	42.22 ±5.88	50.79 ±1.24	52.22 ±4.84
GiGPO	42.46 ±4.47	43.70 ±5.13	45.44 ±3.87	47.41 ±7.40	55.16 ±0.91	57.04 ±5.13
InfoPO	50.40 ±4.47	55.56 ±5.88	55.16 ±0.34	58.52 ±5.70	74.60 ±10.07	74.81 ±9.71
WorldEvolver	50.40 ±2.68	62.59 ±3.21	53.37 ±0.69	64.07 ±4.63	63.10 ±1.19	74.07 ±3.21
PoS / EnvProbe	72.42 ±0.34	70.00 ±2.94	80.56 ±0.91	81.85 ±2.57	**97.22** ±1.50	**97.04** ±0.64
Step-DPO	79.37 ±10.59	75.93 ±10.68	**83.53** ±9.42	79.63 ±9.71	96.63 ±5.84	94.81 ±5.25
Outcome-PG	41.07 ±2.15	44.07 ±4.49	43.06 ±2.25	48.89 ±9.62	56.94 ±4.96	57.41 ±4.49
Outcome-Value	42.26 ±3.90	45.93 ±6.12	45.63 ±3.28	51.48 ±8.04	57.34 ±4.51	57.41 ±4.49
Outcome-RLOO	41.87 ±3.38	44.44 ±3.85	43.85 ±3.96	49.26 ±8.98	56.35 ±5.72	57.41 ±4.49
Info-RL Neighbor	46.23 ±4.81	51.48 ±8.19	50.40 ±3.00	54.44 ±6.76	65.08 ±5.78	59.63 ±5.48
Unstratified	47.62 ±3.31	54.81 ±5.13	52.98 ±1.57	55.93 ±5.25	66.27 ±2.09	67.04 ±5.48
Unweighted	**88.49** ±19.93	**90.37** ±11.13	77.78 ±19.33	**90.00** ±15.40	96.43 ±6.19	94.81 ±7.06
SPAR	**88.49** ±17.91	**91.48** ±6.42	**90.28** ±16.84	**98.15** ±1.28	**99.60** ±0.69	**97.78** ±1.92